USE OF DEEP LEARNING IN HEALTHCARE SYSTEMS FOR NETWORK ATTACK DETECTION AND MITIGATION: A COMPREHENSIVE LITERATURE REVIEW

Gimhana Himasara Thennakoon (IT23623972)
Department of Computing
tmgimhanahimasara@gmail.com

Abstract—The integration of Internet of Medical Things (IoMT) in smart health care has resulted in better patient monitoring, but this has also led to an increase in the network attack surface, making smart health care more vulnerable to complex cyber attacks on its critical infrastructure [1], [2]. The effectiveness of traditional security methods that rely on signatures has been diminishing in recent times due to the rise of novel exploits and polymorphic malware [3], [4]. This review aims to discuss the effectiveness of various Deep Learning (DL) architectures, such as Convolutional Neural Networks (CNN), Long Short-Term Memory Networks (LSTM), and attention mechanisms, in achieving high accuracy in detecting cyber attacks [5], [6]. It also aims to discuss the integration of DL with other paradigms, such as Software Defined Networking (SDN) and Federated Learning (FL), in mitigating cyber attacks in real-time [7], [8], even if hybrid architectures are able to achieve accuracy rates of over 99% [9], [10], [11], [13].

Index Terms—Artificial Intelligence, Blockchain, Cybersecurity, Deep Learning, Distributed Denial of Service (DDoS), Federated Learning, Healthcare Security, Internet of Medical Things (IoMT), Intrusion Detection Systems (IDS), Network Attack Mitigation, Software-Defined Networking (SDN).

I. INTRODUCTION

The transition to Healthcare 4.0 and Internet of Medical Things (IoMT) has revolutionized patient care with connected devices and cloud-based analytical tools [1], [4]. With 75 billion Internet of Things (IoT) devices expected to be operational by 2025, the healthcare domain has been identified as a prime target for cyber attackers due to the extreme value of patient records [4], [5], [3].This has led to the development of many vulnerabilities, which have resulted in unprecedented data breach expenses and patient safety concerns [1], [6], [8]. Distributed Denial of Service (DDoS) and Ransomware attacks, including the recent WannaCry outbreak, have shown their capacity to disable hospital servers and affect life-critical devices such as infusion pumps and robots during surgeries [10], [11], [7].

This complex risk environment includes a wide range of stakeholders, from vulnerable patients to medical device manufacturers, who are under scrutiny regarding the security of their systems [1], [2]. The challenge of finding a balance between ensuring high levels of data accessibility in emergency care situations and the requirement for robust protection against sophisticated attacks within a number of stringent regulatory environments, such as HIPAA and GDPR, is another notable challenge in this environment [5], [19]. The traditional rule-based approach to security is no longer considered adequate in a world of evolving and sophisticated attacks, such as zero-day attacks, and a high volume of biometric and telemetry data [12], [13]. This has led to a number of advances in security, with Deep Learning (DL) now considered a transformative approach that has the ability to automatically identify complex features in raw network traffic that would otherwise go undetected by other means [9], [14], [16].

II. RESEARCH STATEMENT AND OBJECTIVES

This study provides a systematic review of the existing literature on the application of deep learning technology in the infrastructure of the healthcare system to detect and mitigate network attacks, as discussed in the literature [1], [18]. The motivation

behind conducting the literature study comes from the need to shift the paradigm from conventional security measures to intelligent security measures that can provide security to safety-critical Internet of Medical Things infrastructures, as discussed in the literature [1], [16]. The main objectives of the present study are:

1. To develop a structured taxonomy of the existing deep learning architectures used in the field of intrusion detection in healthcare, with particular emphasis on the relative strengths of the techniques in terms of feature extraction and sequence modeling [9], [11].

2. To identify and analyze the key network-level threats in the smart health domain, including volumetric saturation attacks, cryptographic ransomware, and biometric spoofing attacks [1], [7].

3. To evaluate the efficiency of the incorporation of deep learning with Software Defined Networking (SDN) and Federated Learning (FL) in facilitating real-time response to threats and data collaboration in a manner that is privacy-compliant [15], [17].

4. To examine the existing landscape of benchmark datasets and performance measures used in evaluating the efficacy of intrusion detection systems based on deep learning [1], [19].

5. To emphasize the gap in the existing literature, especially with regard to adversarial robustness and interpretability of neural networks, and to identify the research direction that is particularly relevant to the cybersecurity community [10], [20].

III. REVIEW OF THE LITERATURE

The literature pertaining to the implementation of deep learning in the context of the security of healthcare networks covers a wide range of architectures, attacks, and mitigations [1], [10]. The following section will provide a detailed analysis of the different components involved in the implementation of deep learning in the context of the security of healthcare networks.

A. Analysis of the Healthcare Network Attack Landscape

The primary step in the creation of intelligent detection models involves gaining a thorough understanding of the security threats facing the health ecosystem [1], 6]. Network attacks in the health domain have traditionally been classified based on the CIA security triad [1], 15].

1. **Volumetric and Resource Exhaustion Attacks:** Distributed Denial of Service (DDoS) is a significant threat to the availability of healthcare services [1], [12]. Generally, Distributed Denial of Service is a type of attack where a botnet, which is a collection of compromised devices, launches a malicious attack on a server/device, making it inaccessible [6], [11]. This is particularly true in the clinical arena. To illustrate, in 2021, a 30 Gbps distributed denial-of-service attack resulted in the shutdown of a cryptocurrency exchange service, and a similar attack on a hospital gateway could potentially disable hundreds of bedside monitors [1], [15].The most common deniable vectors of DDoS attacks include SYN flood attacks, UDP flood attacks, and ICMP flood attacks [1], [12]. In addition, the incorporation of light medical sensor protocols such as MQTT and CoAP has made it easy for attackers to launch attacks such as MQTT Packet Crafting, which threatens critical healthcare applications by overloading connections [1], [6].
2. **Biometric Spoofing and Man-in-the-Middle (MitM) Attacks:** The risks to data integrity and confidentiality are high in the case of MitM attacks that involve eavesdropping on the communication between the medical sensors and the central hub or altering the data that is transmitted in the process [1], [13]. Spoofing, however, is characterized by the impersonation of an authentic medical device, such as the pacemaker of a patient's heart or the insulin pump used to regulate the levels of insulin in the body, with the aim of sending false information [1], [13], [21]. Research has also indicated the potential to leverage common medical devices, such as speakers or infrared lasers, to carry out spoofing attacks on medical monitoring devices such as Fitbits and insulin pumps, with the potential to lead to fatal outcomes [1], [10].

3. **Ransomware and Advanced Malware in the Healthcare Industry:** Ransomware has also become a notable concern in the healthcare industry, where regional studies have shown a twofold increase in ransomware attacks on hospitals during the past five years [15], [7]. Ransomware attacks often involve the encryption of clinical information and electronic health records, where attackers demand a ransom for providing the decryption key [1], [8]. The WannaCry ransomware attacks of 2017 can be taken as a prototypical case study with 200,000 endpoints affected worldwide and emergency departments in the United Kingdom being temporarily shut down [1], [7]. Other than the use of encryption in ransomware, other types of malware, such as PowerGhost and Petya, make use of hospital networks to disseminate the infection, which in turn affects pharmacy and radiology services [1], [17].

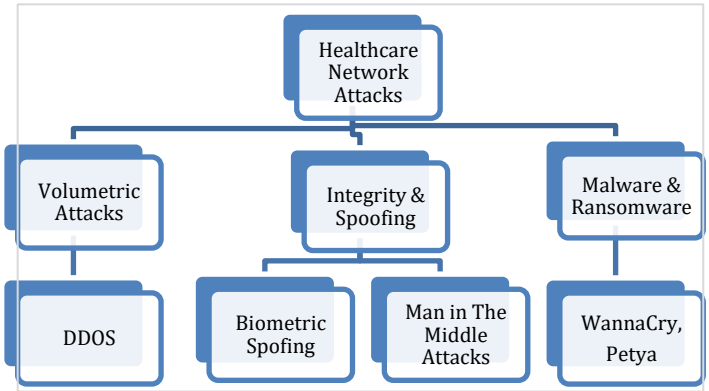

Figure 2 Taxonomy of Healthcare Network Attacks

B. Deep Learning Architectures for Intrusion Detection

The major advantage that deep learning methods possess over regular statistical methods is related to the hierarchy that these methods possess for acquiring feature representations from high-dimensional data [1], [9]. In healthcare scenarios, where data might be sparse and noisy, acquiring feature representations is vital for ensuring security [1] [16].

1. **Supervised Learning of Instance Level Patterns (DNN, CNN):** Recently, deep neural networks with several hidden layers have shown significant effectiveness in modeling complex and nonlinear relationships commonly encountered in network flow data [3], [19]. From the empirical evaluation of DNNs, it is seen that there is a 15% improvement in accuracy over traditional machine learning approaches, which are Decision Trees and Naive Bayes, for benchmark datasets [14], [3]. The current trend in network security suggests an increasing trend in the adoption of Convolutional Neural Networks (CNNs) [1], [12]. Although CNNs are typically used in image processing, they are equally useful in network data represented in 1D or 2D vectors. The architecture of the CNN has been proven successful in the extraction of spatial features from the network data, thus reducing the dimensions and increasing computational efficiency [1], [13]. A good example is the optimization of the 1D CNN architecture to achieve 99% accuracy on the CICIOMT2024 dataset for the purpose of effectively discriminating 18 distinct categories of attacks [1], [16].

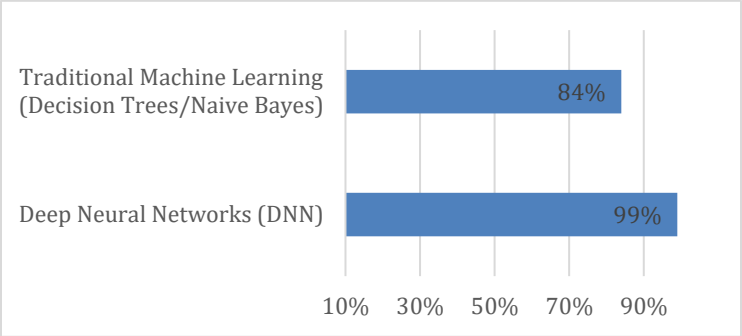

Figure 1 Accuracy Comparison of Traditional Machine Learning vs. Deep Neural Networks

2. **Instance Learning (DNN and CNN):** Deep Neural Networks (DNNs) with several hidden layers can effectively model complex non-linear relationships that often occur in network flow data, as reported in [3], [19]. Experimental results also demonstrate that DNNs can achieve higher accuracy than other machine learning algorithms like Decision Trees and Naive Bayes by up to 15%, as reported in [3], [14].Recent literature and existing trends suggest a preference for Convolutional Neural Networks (CNNs) in the area of network security [1], [12]. Although CNNs are primarily developed for

processing images, they can apparently support network traffic data in both vector and image forms [1], [19]. Convolutional Neural Networks have been proven efficient in feature and spatial pattern learning with the help of convolution and pooling layers. The efficiency of CNNs is due to the dimensionality reduction feature and improved computational performance achieved with the help of these layers [1], [13].For example, a one-dimensional convolutional neural network (1D-CNN) model, which was optimized for time series data, was able to attain a 99% accuracy rate in distinguishing between 18 different types of attack modes using the CICIOMT2024 dataset [1], [16].

3. **Supervised Sequence Learning using RNN and LSTM:** The sequential properties in network traffic data are innate, and the sequence of packets contains more information than individual packets themselves [1], [20]. Recurrent Neural Networks are designed to handle sequential data and can learn from the sequence using their internal state [1], [9].However, recurrent neural networks have limitations in learning long-term dependencies because of the problem of vanishing gradients [1], [20]. On the other hand, Long Short-Term Memory is a type of neural network for processing sequential data and is capable of learning long-term dependencies by overcoming the problem of vanishing gradients [1], [20]. Bidirectional LSTM networks are also highly efficient in handling sequence information and can provide contextual information necessary for identifying slight anomalies such as low-rate DDoS attacks [2], [10].
4. **Hybrid architectures and attention mechanisms** are found to be the most powerful paradigms in recent literature, which again stresses the unification of diverse methodological strengths [5], [21]. Specifically, the CNN-LSTM architecture combines the capabilities of convolutional neural network layers for spatial feature extraction with the capabilities of long short-term memory layers for temporal behavior modeling [10], [13]. The inclusion of an attention mechanism improves the performance of the architecture as it facilitates the assignment of differential weights to the relative importance of particular time steps in the input sequence [1], [2].Thus, by focusing on the essential characteristics, like the occurrence of sudden changes in the frequency of packets, it is possible for the CNN-BiLSTM-AT model to achieve classification accuracy up to 99.78% [1], [2].

IV. INTEGRATED PARADIGMS FOR ATTACK MITIGATION

Detection is just the beginning of the security equation, and healthcare systems require robust mitigation strategies to contain and respond to these types of threats in real time [15], [18].

A. Software-Defined Networking (SDN) for Containment

This article is an attempt to explain SDN as a new and innovative approach to managing a network, which is enabled by decoupling the control plane from the data plane [11, 15]. With regards to healthcare networks, it is explained that SDN is comprised of a controller that has a holistic, end-to-end view of the entire network [6, 15]. Once a compromised medical device is detected by a deep learning algorithm, a new set of flow rules can be disseminated by the SDN controller to switches that are associated with the compromised device, thereby isolating it from the larger network [6, 16], and thus quarantining the ransomware attack within the hospital's networks [14, 16].

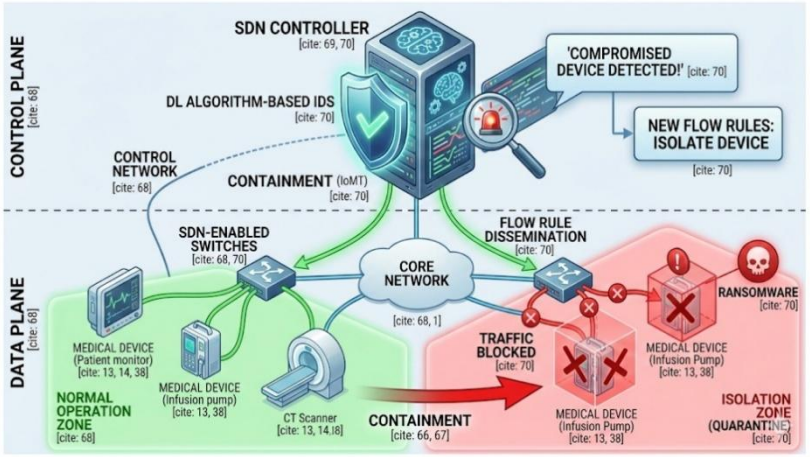

Figure 3 SDN-based mitigation architecture for medical device isolation and network containment.

B. Federated Learning (FL) for Privacy-Preserving Defense

There are many cases in which data in the health domain are distributed or "siloed" across many institutions due to legal or security concerns, such as those addressed in HIPAA [1], [17]. Federated Learning allows many hospitals to jointly construct a single intrusion detection model without sharing their original patient data [5], [16].The hospitals train independently a model using individual data sets and share only the learned weights with a central server that aggregates these contributions to produce an improved global model [1], [17]. Even advanced security measures like Distributed Selective Stochastic Gradient Descent (DSSGD) can control the probability of an attempted model inversion attack, in which an attacker seeks to recover the original information from the shared weights [1], [16].

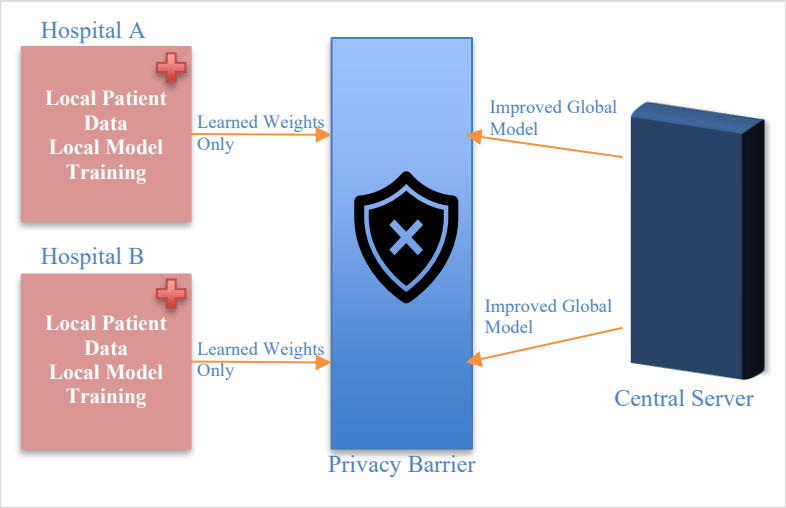

Figure 4 Hospitals independently train models using individual datasets and share only the learned weights with the central server, ensuring data privacy

V. EVALUATION OF BENCHMARK DATASETS AND METRICS

The validity of research in this particular domain is highly dependent on the quality and relevance of the datasets used for training and testing deep learning models [1], [19].

A. Traditional versus Contemporary Datasets: Traditionally, research methodologies employed the KDDCUP99 and NSL-KDD datasets. Currently, these datasets are considered outdated, as they are not effective in addressing the complexity of modern-day network traffic and attack patterns [1], [19]. Contrarily, modern research methodologies employ contemporary and practical datasets such as the CICIDS2017/2018, CICDDoS2019, and the advanced CICIoMT2024 datasets [1], [16].

B. Performance Evaluation Metrics: To obtain a complete picture of the performance of the model, several metrics have been used by researchers [1], [19]. Though accuracy is used as a standard performance measure, it might not be a reliable one in situations where 99% of the traffic is benign [1], [14]. Therefore, precision, recall, F1-score, and ROC-AUC are used.

VI. FUTURE RESEARCH

Deep learning has significantly improved the cybersecurity of healthcare. However, there are several important domains where new and emergent technologies have been required [8], [22].

A. Adversarial Robustness and the Security of AI

One of the major gaps in the literature is the lack of consideration of the vulnerability of the deep learning models themselves [8], [9]. Adversarial attacks, in this context, refer to methods by which an adversary seeks to trick the neural network by making small modifications to the input of the network [8], [9]. In the future, there should be emphasis on adversarial training and defensive distillation as methods for creating models with adversarial robustness [8], [20] .

B. Real-Time Self-Healing and Autonomous Mitigation

The current mitigation mechanisms are mostly reactive in nature, with a detection-response cycle involving a long latency period [1], [22]. There is a pressing need to explore autonomous self-healing

networks using DRL [22]. Such networks are required to be self-healing in nature, without human intervention [1], [22].

VII. CONCLUSION

Deep learning has firmly secured its place as a key component in modern cybersecurity, utilizing the dynamic and flexible capabilities of the approach to detect ever-evolving and intricate patterns of network-based attacks with unmatched accuracy [1], [9]. From the analysis, it is clear that hybrid-based solutions and privacy-centric collaborative techniques such as Federated Learning hold the key to ensuring the security of clinical information [16], [17]. However, the unexplained and unpredictable nature of neural networks and their potential to be compromised by adversarial attacks present two key hurdles in the adoption of deep learning-based solutions in the clinical domain [8], [20]. The future of healthcare cybersecurity suggests the need to develop "lightweight, transparent, and self-healing solutions to protect the network edge against volumetric, biometric, and cryptographic-based threats" [23] [24].

APPENDIX: PERFORMANCE EVALUATION METRICS FOR NIDS

Metric	Formula	Clinical Implication
Accuracy	(TP + TN) / (TP + TN + FP + FN)	General reliability of the IDS.
Precision	TP / (TP + FP)	Minimizes "false alarms" disrupting staff.
Recall (DR)	TP / (TP + FN)	Ensures no malicious attacks are overlooked.
F1-Score	2 * (Prec * Rec) / (Prec + Rec)	Balances trade-offs in imbalanced data.
FAR	FP / (FP + TN)	Rate of benign traffic flagged as attack.

- **True Positive (TP):** The system correctly identifies a malicious activity as an attack.
- **True Negative (TN):** The system correctly identifies legitimate network activity as normal.
- **False Positive (FP):** The system incorrectly identifies legitimate activity as an attack. This could cause unnecessary interruptions in clinical workflow.
- **False Negative (FN):** The system misses a real attack and incorrectly identifies it as normal activity. This is a critical concern in a healthcare environment because it suggests a security threat has gone undetected.

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REFERENCE

[1] Vinayakumar Ravi, Tuan D. Pham, Mamoun Alazab, "Deep learning-based Network Intrusion Detection System for Internet of Medical Things," 2026. [Online]. Available: https://doi.org/10.1109/IOTM.001.2300021.

[2] Mr. Shashi Maurya, Dr. Neha Gupta , Dr. Rahul Kumar, "Deep Models against Deep Threats: A Review of Deep Learning for Cyber Attack Mitigation," 2026. [Online]. Available: https://doi.org/10.47772/IJRISS.2026.10190024.

[3] B. Ta¸sc, "Deep-Learning-Based Approach for IoT Attack and Malware Detection - MDPI," 2024. [Online]. Available: https://doi.org/10.3390/app14188505.

[4] Mourad Benmalek, Abdessamed Seddiki,Kamel-Dine Haouam, "SNN-IoMT: A Novel AI-Driven Model for Intrusion Detection in Internet of Medical Things," 2025. [Online]. Available: https://doi.org/10.32604/cmes.2025.062841.

[5] Muhammad Babar,uhammad Usman Tariq, Basit Qureshi, Zabeeh Ullah, Fahim Arif, Zahid Khan, "An Efficient and Hybrid Deep Learning-Driven Model to Enhance Security and Performance of Healthcare Internet of Things," 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3536638.

[6] D. MaheshT,Sampath Kumar, "OptCosineDCNN: An Advanced Network Attack Detection and Mitigation in SDN Environments," 2025. [Online]. Available: https://doi.org/10.1080/01969722.2025.2552941.

[7] Abdullah Ahmed Bahashwan ,MohammedAnbar,Selvakumar Manickam,Taief Alaa Al-Amiedy,MohammadAdnanAladaileh,IznanH.Hasbullah, "A Systematic Literature Review on Machine Learning and Deep Learning Approaches for Detecting DDoS Attacks in Software-Defined Networking," 2023. [Online]. Available: https://doi.org/10.3390/s23094441.

[8] Gianmarco Di Palma , Roberto Scendoni , Davide Ferorelli, Anna De Benedictis,Vittoradolfo Tambone, Francesco De Micco, "AI-Induced Cybersecurity Risks in Healthcare:A Narrative Review of Blockchain-Based SolutionsWithin a Clinical Risk Management Framework," 2025. [Online]. Available: https://doi.org/10.2147/RMHP.S544523.

[9] J Aruna SanthiG ,Vijaya Saradhi, "Attack detection in medical Internet of things using optimized deep learning: enhanced security in healthcare sector," 2021. [Online]. Available: https://doi.org/10.1108/DTA-10-2020-0239.

[10] Nithesh Naik, Neha Surendranath, Sai Annamaiah Basava Raju, Chennaiah Madduri,Nagaraju Dasari, Vinod Kumar Shukla,Vathsala Patil, "Hybrid deep learning-enabled framework for enhancing security, data integrity, and operational performance in Healthcare Internet of Things (H-IoT)," 2025. [Online]. Available: https://doi.org/10.1038/s41598-025-15292-2.

[11] Erikson J. de Aguiar, Caetano Traina Jr., Agma J. M. Traina, "Security and Privacy in Machine Learning for Health Systems: Strategies and Challenges," 2023. [Online]. Available: https://doi.org/10.1055/s-0043-1768731.

[12] MOZAMEL M. SAEED, "An AI-Driven Cybersecurity Framework for IoT: Integrating LSTM-Based Anomaly Detection," 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3576506.

[13] K Vaisakhkrishnana, Gadde Ashoka, Parimarjan Mishraa, T.Gireesh Kumara, "Guarding Digital Health Deep Learning for Attack Detection in Medical IoT," 2024. [Online]. Available: https://doi.org/10.1016/j.procs.2024.04.235.

[14] R. Alluhaibi, "Quantum resistant blockchain and deep learning revolutionize secure communications for autonomous vehicles," 2026. [Online]. Available: https://doi.org/10.1038/s41598-025-28938-y.

[15] Sidra Batool,Muhammad Aslam,Edore Akpokodje,Syeda Fizzah Jilani, "A Comprehensive Review of DDoS Detection and Mitigation in SDN Environments: Machine Learning, Deep Learning, and Federated Learning Perspectives," 2025. [Online]. Available: https://doi.org/10.3390/electronics14214222.

[16] Kaoutar Tebbaa,Oumaima Chakir,Yassine Maleh,Mustapha Belaissaoui, "Mitigating DDoS attacks in software-defined networks: a systematic literature review," 2025. [Online]. Available: https://doi.org/10.1007/s42044-025-00386-x.

[17] Ashfaq Ahmad Najara, S. Manohar Naik, "DDoS attack detection using CNN-BiLSTM with attention mechanism," 2025. [Online]. Available: https://doi.org/10.1016/j.teler.2025.100211.

[18] JAMIL ISPAHANY,MD. RAFIQUL ISLAM,MD. ZAHIDUL ISLAM,M. ARIF KHAN, "Hybrid deep learning-enabled framework for enhancing security, data integrity, and operational performance in Healthcare Internet of Things (H-IoT) environments," 2024. [Online]. Available: https://doi.org/10.1038/s41598-025-15292-2.

[19] Hannelore Sebestyen, Daniela Elena Popescu,Rodica Doina Zmaranda, "A Literature Review on Security in the Internet of Things: Identifying and Analysing Critical Categories," 2025. [Online]. Available: https://doi.org/10.3390/computers14020061.

[20] Sebastian Ignacio Berrios Vasquez,Pamela Alejandra Hermosilla Monckton,Dante Ivan Leiva Muñoz,Hector Allende, "Zero-Day Threat Mitigation via Deep Learning in Cloud Environments," 2025. [Online]. Available: https://doi.org/10.3390/app15147885.

[21] Mohanad Faeq Ali,Mohammed Shakir Mohmood,Ban Salman Shukur,Rex Bacarra,Jamil Abedalrahim Jamil Alsayaydeh ,Masrullizam Mat Ibrahim,Safarudin Gazali Herawan, "HCAP: Hybrid cyber attack prediction model for securing healthcare applications," 2025. [Online]. Available: https://doi.org/10.1371/journal.pone.0321941.

[22] Naveed Ahmed,Asri bin Ngadi, Johan Mohamad Sharif,

Saddam Hussain , Mueen Uddin, Muhammad Siraj Rathore, Jawaid Iqbal, Maha Abdelhaq , Raed Alsaqour, Syed Sajid Ullah, Fatima Tul Zuhra, "Network Threat Detection Using Machine/Deep Learning in SDN-Based Platforms: A Comprehensive Analysis of State-of-the-Art Solutions, Discussion, Challenges, and Future Research Direction," 2022. [Online]. Available: https://doi.org/10.3390/s22207896.

[23] Sulaiman Alamro, "AnomLocal: A hybrid local-global anomaly detection model for network security using federated learning," 2026. [Online]. Available: https://doi.org/10.1371/journal.pone.0339981.

[24] Lekhetho Joseph Mpoporo, Pius Adewale Owolawi,Chunling Tu, "Deep Reinforcement Learning Algorithms for Intrusion Detection: A Bibliometric Analysis and Systematic Review," 2026. [Online]. Available: https://doi.org/10.3390/app16021048.

AUTHOR PROFILE

Gimhana Himasara Tennakoon is an undergraduate student currently studying towards a Bachelor of Computing degree with a specialization in Cybersecurity at the Sri Lanka Institute of Information Technology (SLIIT) in Kandy, Sri Lanka. His high school education focused on Information & Communication Technology, Physics, and Combined Mathematics, making his educational background fall within the physical science and information communication domain.

The student researcher is currently focused on rethinking security paradigms from traditional security to intelligent security for safety-critical Internet of Medical Things (IoMT) infrastructures. His current research focuses on the efficacy of different Deep Learning (DL) models, such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models, for the detection of sophisticated cyber-attacks. Current research areas of interest include network attack mitigation techniques, Software-Defined Networking (SDN), and the incorporation of Federated Learning for the purpose of providing privacy-preserving security solutions.